

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

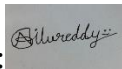
Total Marks: 25

Code of Conduct

I G. Venkata Niranjan Reddy (192525199) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Assessment Tasks

Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.
2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy. Design a solution using cloud-native backup, replication, and versioning. Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved. Provide steps to ensure business continuity during failures.
4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.
5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

1. ANSWER:

A media company can design a scalable cloud architecture using cloud storage, CDN, load balancing, and auto-scaling. Video files can be stored in cloud object storage, while a Content Delivery Network (CDN) caches popular videos at edge locations close to users. This reduces latency and buffering because users can access videos from nearby servers. A load balancer distributes incoming requests among multiple application servers, ensuring that no single server becomes overloaded. Auto-scaling can automatically increase the number of servers when millions of users access the platform during peak hours and reduce them when traffic decreases. For cost optimization, the company can use pay-as-you-go resources, auto-scaling, storage lifecycle policies, CDN caching, and reserved pricing for predictable workloads. This architecture provides high performance, scalability, reliability, and cost efficiency.

Scalability and Caching

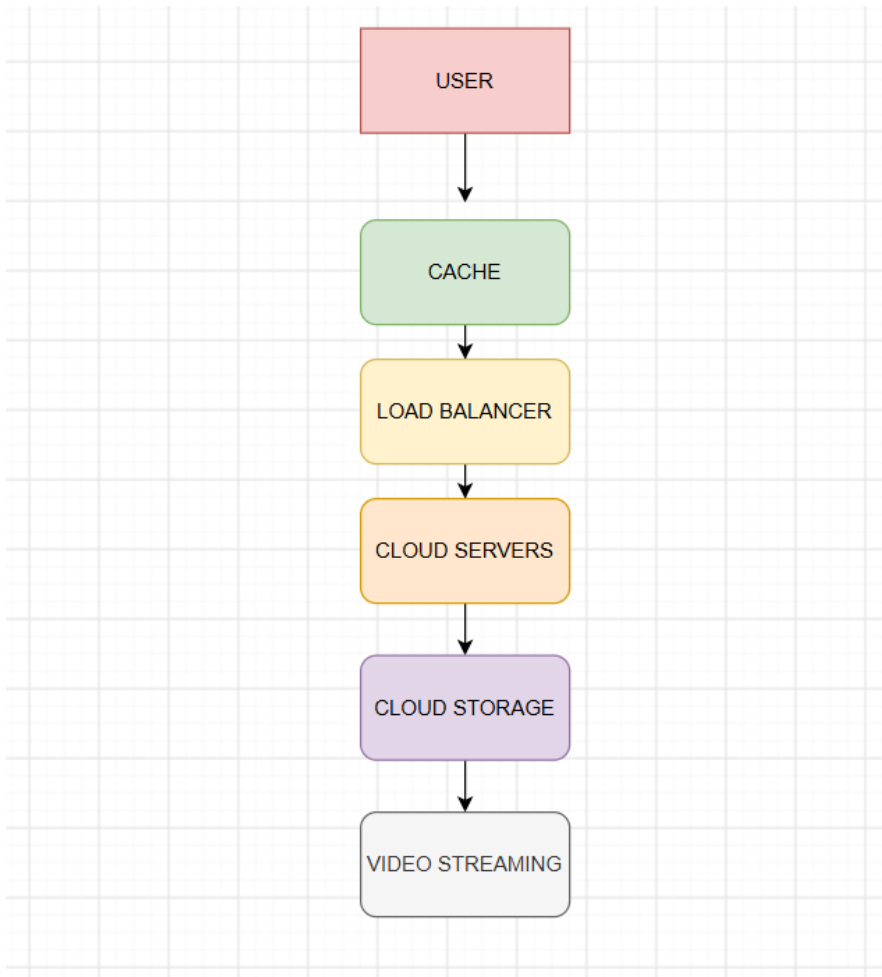
- **CDN:** Delivers videos from locations close to users.
- **Caching:** Frequently watched videos are stored temporarily at edge servers.
- **Cloud Storage:** Provides scalable and durable storage for large video files.
- **Adaptive Streaming:** Automatically adjusts video quality according to network speed.
- **Database Scaling:** Read replicas can handle a large number of user requests.

Load Balancing and Auto-Scaling

A load balancer distributes incoming user requests among multiple application servers. If one server fails, requests are redirected to healthy servers. Auto-scaling automatically adds new servers when the number of users increases and removes unnecessary servers when demand decreases. This ensures that the platform can handle millions of users without overloading individual servers.

Cost Optimization

1. It uses CDN caching to reduce bandwidth usage.
2. It uses auto-scaling to avoid paying for unused servers.
3. It uses cheaper storage for old and rarely accessed videos.



2.ANSWER:

A fintech startup can use containers and Kubernetes to deploy its applications consistently across multiple cloud providers. Applications can be packaged as Docker containers and managed using Kubernetes clusters in different cloud environments.

Kubernetes provides automated deployment, container management, service discovery, load balancing, and scaling. Kubernetes Services and DNS can help applications discover and communicate with each other without depending on fixed IP addresses.

Proposed Solution

- Package applications as Docker containers.
- Use Kubernetes as the container orchestration platform.
- Use the same container images and deployment configurations across all environments.

- Use Infrastructure as Code (IaC) tools such as Terraform to maintain consistent infrastructure.

Service Discovery

- Kubernetes provides built-in service discovery using Services and DNS.
- Applications can communicate using service names instead of fixed IP addresses.

Benefits of Multi -Cloud

- Reduces dependency on a single cloud provider.
- Improves availability and disaster recovery.
- Allows the company to select the best services from different providers.
- Provides flexibility in managing workloads and costs.

Risks of Multi -Cloud

- Increased infrastructure complexity.
- Higher monitoring and management requirements.
- Different cloud APIs and services may create compatibility problems.

3.ANSWER:

An e-learning platform should implement a cloud-native backup and disaster recovery strategy to protect its data from accidental deletion and system failures. Important databases and files should be backed up automatically at regular intervals, while object storage versioning can preserve previous versions of deleted or modified files. Critical data should also be replicated to another availability zone or geographic region so that it can be recovered if the primary environment fails.

Proposed Solution

- Store application data in highly durable cloud databases.
- Enable database backups and point-in-time recovery.
- Enable object versioning for files and learning materials.
- Replicate critical data to another availability zone or region.

- Use automated backup policies and lifecycle management.

Backup, Replication, and Versioning

- Backup: Creates copies of databases and application data.
- Replication: Maintains copies of data in another location for disaster recovery.
- Versioning: Keeps previous versions of deleted or modified files.
- Immutable backups: Prevent backups from being accidentally or maliciously deleted.

RTO and RPO:

- RTO (Recovery Time Objective): The maximum acceptable time to restore the service.
- RPO (Recovery Point Objective): The maximum acceptable amount of data loss measured in time.

4.ANSWER:

A government agency can use a hybrid cloud architecture to keep sensitive and compliance-related applications on-premises while deploying non-sensitive applications in the public cloud. The on-premises environment can store confidential citizen information and critical government databases, while the public cloud can host websites, public services, and scalable non-sensitive applications. Secure communication between the two environments can be established using VPN connections or dedicated private network connections.

Proposed Architecture

On-Premises Data Center

- Sensitive citizen data.
- Government databases.
- Internal systems.

Public Cloud

- Public websites.
- Non-sensitive applications.
- Analytics and scalable workloads.

Secure Communication

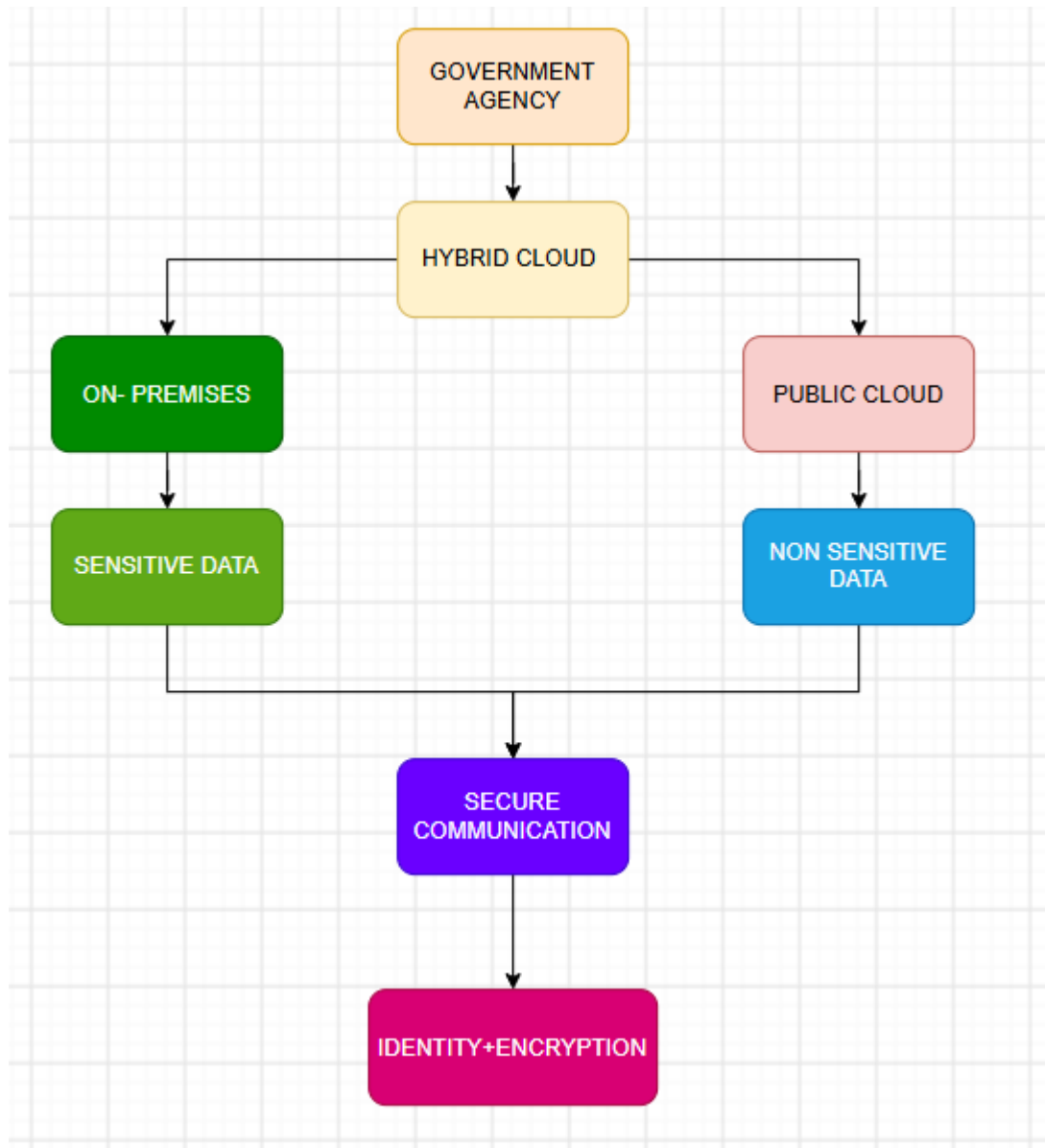
- Use encrypted VPN tunnels or dedicated private connectivity.
- Use firewalls and network security groups.
- Separate workloads using network segmentation.

Identity Management

- Implement centralized identity and access management.
- Use Single Sign-On (SSO) for authorized users.
- Apply Role-Based Access Control (RBAC).
- Use Multi-Factor Authentication (MFA).

Encryption

- Encrypt data **at rest** using cloud and on-premises encryption mechanisms.
- Encrypt data **in transit** using TLS.
- Use centralized key management systems.
- Rotate encryption keys regularly.



5.ANSWER:

A SaaS provider that wants its application to remain operational even when an entire cloud region becomes unavailable should implement a multi-region architecture. In this design, the application is deployed in at least two geographically separate cloud regions. Each region can contain multiple availability zones, load balancers, application servers, databases, and supporting services. A global DNS service or traffic management system can distribute users between regions. If one region becomes unavailable, traffic can automatically be redirected to the healthy region. This approach reduces downtime and improves the overall availability of the SaaS application.

Proposed Multi-Region Architecture

Users → Global DNS/Traffic Manager → Region 1 / Region 2 → Load Balancers → Application Servers

- Deploy the application in at least two geographically separate regions.
- Use load balancers in each region.
- Run multiple application instances across different availability zones.

Data Replication

- Use database replication between regions.
- For critical applications, use multi-region or globally distributed databases.
- Replicate object storage to a secondary region.
- Select synchronous replication when strong consistency is required and latency allows it.

Monitoring and Alerting

Monitor:

1. Application availability.
2. CPU and memory usage.
3. Network traffic.
4. Database performance.

5. Error rates.

Alerts

1. Server or region failures.
2. High error rates.
3. Increased latency.
4. Database replication failures.

Proactive Detection

- Use centralized logging and monitoring.
- Create dashboards for real-time system health.
- Use automated health checks.
- Apply anomaly detection where appropriate.

